

- ① When a 5×10^{-9} test charge is placed at a point, it experiences a force of $2 \times 10^{-4} \text{ N}$ in the x-direction that is electric field at that point.

Here, $F = 2 \times 10^{-4} \text{ N}$

$$q = 5 \times 10^{-9} \text{ C}$$

We know that, $F = qE$

$$\Rightarrow E = \frac{F}{q}$$

$$= \frac{2 \times 10^{-4}}{5 \times 10^{-9}}$$

$$= 4 \times 10^4 \text{ N C}^{-1}$$

(Ans)

- ② What is the force on an electron placed at the point where the electric field intensity is $4 \times 10^4 \text{ N C}^{-1}$.

Here, $E = 4 \times 10^4 \text{ N C}^{-1}$

$$q = 1.6 \times 10^{-19} \text{ C (electron)}$$

We know that,

$$F = qE$$

$$= (1.6 \times 10^{-19}) \times (4 \times 10^4)$$

$$= 6.4 \times 10^{-23} \text{ N}$$

For electron, $F = -6.4 \times 10^{-23} \text{ N}$.

(Ans)

③ Calculate the field E due to a dipole moment 4.5×10^{-10} C/m at a distance 1m from it on the perpendicular bisector.

Given that,

$$\text{dipole moment, } p = 4.5 \times 10^{-10} \text{ C/m}$$
$$r = 1 \text{ m}$$

We know that,

$$E = \frac{p}{4\pi\epsilon_0} \cdot \frac{1}{r^3}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{r^3}$$

$$= (9 \times 10^9) \times \frac{(4.5 \times 10^{-10})}{1^3}$$

$$= 4.05 \text{ V/m} \quad \text{(Ans)}$$

Q) What is the potential at the surface of an Aluminium nucleus $r = 4.5 \times 10^{-15} \text{ cm}$ and it's charge $q = 13e$.

Given that,

$$r = 4.5 \times 10^{-15} \text{ cm}$$

$$\text{Charge, } q = 13 \times (-1.6 \times 10^{-19}) \\ = -2.08 \times 10^{-18} \text{ C}$$

Potential, $V = ?$

$$\text{charge density, } \sigma = \frac{\phi}{A} \\ = \frac{\phi}{4\pi r^2}$$

We know that,

$$E = \frac{V}{r} \quad \text{--- (I)}$$

$$E = \frac{\sigma}{\epsilon_0} \quad \text{--- (II)}$$

Equating (I) and (II) $\Rightarrow$

$$\frac{V}{r} = \frac{\sigma}{\epsilon_0}$$

$$\text{or, } \frac{V}{r} = \frac{\phi}{\epsilon_0 4\pi r^2}$$

$$\text{or, } V = \frac{1}{4\pi\epsilon_0} \cdot \frac{\phi}{r}$$

$$\text{or, } V = (9 \times 10^9) \times \frac{2.08 \times 10^{-18}}{4.5 \times 10^{-15}}$$

$$\therefore V = 4.16 \times 10^6 \text{ V}$$

(Ans)

- ⑤ Two equal charges of magnitude $2 \times 10^{-6} \text{ C}$ are placed at a distance 8 cm from each other. Find the magnitude of electric intensity at point 3 cm from the midpoint of the line to the two charges along the perpendicular bisector of the line.

Given that, $q = 2 \times 10^{-6} \text{ C}$

$$2a = 8 \text{ cm} = 0.8 \text{ m}$$

$$a = 4 \text{ cm} = 0.4 \text{ m}$$

$$r = 3 \text{ cm}$$

$$= 0.3 \text{ m}$$

$$\frac{1}{4\pi\epsilon_0} = (9 \times 10^9)$$

$$E = ?$$

We know that,

$$\begin{aligned} E &= \frac{2aq}{4\pi\epsilon_0} \times \frac{1}{(a^2+b^2)^{3/2}} \\ &= \frac{1}{4\pi\epsilon_0} \times \frac{2aq}{(a^2+b^2)^{3/2}} \\ &= (9 \times 10^9) \times \frac{0.8 \times (2 \times 10^{-6})}{\{(0.4)^2 + (0.3)^2\}^{3/2}} \\ &= 9 \times 10^9 \times 1.28 \times 10^{-5} \\ &= 1.152 \times 10^5 \text{ N/C} \end{aligned}$$

(Ans)

⑥ An electric dipole consists of two opposite charges of magnitude $q = 2 \times 10^{-6} \text{ C}$ separated by 1 cm. The dipole is placed in an external field $2 \times 10^5 \text{ N/C}$. Calculate (1) The maximum torque on the dipole (2) Work done to turn the dipole through 180° starting from a position of $\theta = 0^\circ$

Given that,

$$\begin{aligned} p &= q \times r = 2 \times 10^{-6} \times 0.01 \\ &= 2 \times 10^{-8} \text{ m} \end{aligned}$$

$$E = 2 \times 10^5 \text{ N/C}$$

$$\text{Again, } p = 2 \times 10^{-8}$$

$$E = 2 \times 10^5 \text{ N/C}$$

$$W = ?$$

(i) We know that,

$$\begin{aligned}\tau &= PE \sin \theta \\ &= 2 \times 10^{-12} \times 2 \times 10^5 \times \sin 90^\circ \\ &= 4 \times 10^{-4} \text{ Nm}\end{aligned}$$

(ii) We know that,

$$\begin{aligned}W &= \int_0^{180} PE \sin \theta \cdot d\theta \\ &= -PE [\cos \theta]_0^{180} \\ &= -PE (\cos 180 - \cos 0) \\ &= -PE (-2) \\ &= 2PE \\ &= 2 \times 2 \times 10^{-12} \times 2 \times 10^5 \\ &= 8 \times 10^{-2} \text{ J}.\end{aligned}$$

(Ans)

⑦ A 5 coulombs charge is placed on the circumference of a circle of radius 0.5 m. Find the potential and intensity at a distance 25 cm from the centre.

① We know that,

$$V = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r}$$

$$= (9 \times 10^9) \times \frac{5}{0.5}$$

$$= 9 \times 10^9 \text{ V}$$

② Electric field intensity

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

$$= (9 \times 10^9) \times \frac{5}{(25 \times 10^{-2})^2}$$

$$= 7.2 \times 10^{11} \text{ N/C}$$

Given that,

$$q = 5 \text{ C}$$

$$r = 0.5 \text{ m}$$

$$\text{Potential } V = ?$$

$$q = 5 \text{ C}$$

$$r = 25 \times 10^{-2}$$

$$E = ?$$

⑧ What is the electric potential at the surface of a gold nucleus. The radius is $6.6 \times 10^{-15} \text{ m}$ and the atomic number is 79.

We know that,

$$V = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r}$$

$$= (9 \times 10^9) \times \frac{1.264 \times 10^{-17}}{6.6 \times 10^{-15}}$$

$$= 1.723 \times 10^{-3} \text{ V}$$

Given that,

$$r = 6.6 \times 10^{-15}$$

Atomic number $Z = 79$

$$V = ?$$

Charge, $q = Ze$

$$= 79 \times 1.6 \times 10^{-19}$$

$$= 1.264 \times 10^{-17} \text{ C}$$

⑨ At one time the positive charge in the atom was thought to be distributed uniformly throughout a sphere with a radius of about $1 \times 10^{-10} \text{ m}$. That is throughout the entire atom. Calculate the electric field strength at the surface of a gold atom ($Z = 79$) on this supposition.

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r^2}$$

$$= (9 \times 10^9) \times \frac{(1.264 \times 10^{-17})}{(1 \times 10^{-10})^2}$$

$$= 1.1376 \times 10^{13} \text{ N/C}$$

Given that,

$$r = 1 \times 10^{-10} \text{ m}$$

$$Z = 79$$

$$q = Ze = 1.264 \times 10^{-17} \text{ C}$$

$$E = ?$$

⑩ A negative point charge of 10^{-6} C is distributed in air of the origin of a rectangular co-ordinate system. A second negative point charge of 10^{-4} C is situated on the positive x-axis at a distance of 50 cm from the origin. What is the force on the second charge.

$\vec{F} = \frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{r^2} \times \hat{r}$ $= (9 \times 10^9) \times \frac{(-10^{-6})(-10^{-4})}{(50 \times 10^{-2})^2}$ $F = 3.6 \text{ N}$ $\therefore \vec{F} = 3.6 \text{ N } (+i)$	Given that, $q_1 = -10^{-6} \text{ C}$ $q_2 = -10^{-4} \text{ C}$ $r = 50 \times 10^{-2} \text{ m}$ $F = ?$
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